TheAgentCompany failure analysis: English vs Bangla (OpenHands with gemma4:31b)

Generated by `analysis/tac/fa_report.py` on 2026-10-10, following `failure_analysis_pipeline.md` step by step. Every number below is recomputed from `analysis/tac/tac_runs.pkl` (built from the evaluator outputs in `TheAgentCompany/evaluation/outputs*`) and the 8 case notes in `tac_trace_notes.xlsx`.

Summary of results

- **English ≥ Bangla on every flipped task, but the pilot is too small to test it.** Full completion 39.1% in English against 26.1% in Bangla on 23 task-runs (gap +13.0 pp, 95% interval [0.0, 26.1] pp, McNemar p = 0.250); 3 tasks pass only in English, 0 only in Bangla.
- **The only stable English-pass/Bangla-fail case is a checker artifact:** in admin-ask-for-meeting-feedback the simulated coworkers answer in Bangla and the checker needs the English word "helpful"; the agent's steps match English in both Bangla runs.
- **Run-to-run variation is as large as the language gap.** Re-running Bangla changed full completion on 1 of 13 rerun tasks and the score on 3; choosing the first instead of the latest Bangla run moves the gap from +13.0 pp to +17.4 pp.
- **No genuine Bangla agent failure was found** among the 8 cases read side by side; most failures fail in both languages (14 of 23 task-runs fail in both).
- **Limits.** 21 unique tasks, one model, one run per cell plus some Bangla reruns; single annotator; every claim here is descriptive.

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0. Scope and how the pipeline was adapted

The pipeline was written for tau2-style dialogue benchmarks. This report applies it to **TheAgentCompany (TAC): long-horizon company tasks run by an OpenHands agent with gemma4:31b in a simulated company (RocketChat, GitLab, ownCloud, Plane) with LLM-played coworkers (NPCs)**. Each task has weighted checkpoints; the pipeline's pass/fail is **full completion** (all checkpoint points), partial credit is reported alongside.

Pipeline item	In this report
Dataset	One: TheAgentCompany. Domain = task set: Set 1 (RocketChat and GitLab tasks) and Set 2 (ownCloud tasks).
Model	One: gemma4:31b inside OpenHands 0.42. Model-wise splits are not applicable.
Tasks	23 task-runs per language (21 unique tasks; sde-install-go and sde-install-openjdk appear in both sets). 2 tasks ran only in English (research-answer-questions-on-paper, sde-copy-table-from-pdf-to-xlsx): runtime build stalls.
"Bangla user simulator"	The NPC coworkers play this role. In the Bangla cells they answer in Bangla. C2 is kept as "NPC error".
Bangla reruns	13 tasks were run twice in Bangla (`*_rc_bn_full` , `*_set2_bn_full`). Main analysis uses the latest run; the first run is a sensitivity scenario.
Case selection (§3)	Every EN-pass/BN-fail case, plus every task whose Bangla reruns disagree, plus both-fail tasks whose partial score differs: 8 cases.
Every failed run	Not hand-coded beyond the 8 cases; checkpoint-level outcomes are given for every run (Figure 5).
Second reviewer	**Not done.** All 8 notes are by one annotator (a Claude model).

R. Results of all runs

Run	Task-runs	Full completion	Set 1 (RocketChat, GitLab)	Set 2 (ownCloud)	Mean partial score
English	23	9 / 23 (39.1%)	6 / 11 (54.5%)	3 / 12 (25.0%)	0.515
Bangla (latest run)	23	6 / 23 (26.1%)	5 / 11 (45.5%)	1 / 12 (8.3%)	0.433
Bangla (first run)	23	5 / 23 (21.7%)	4 / 11 (36.4%)	1 / 12 (8.3%)	0.423

R.1 The study in two figures

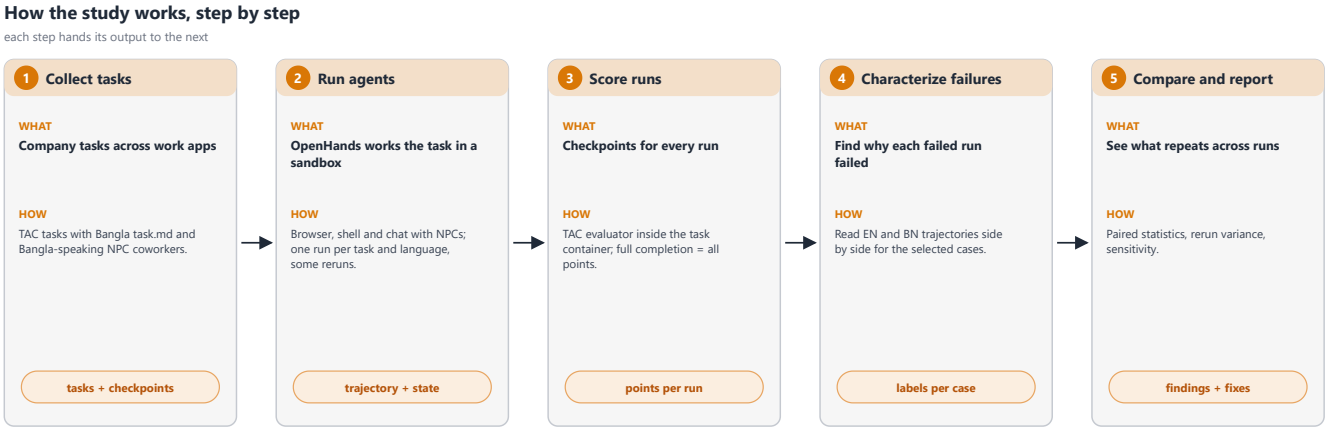


Figure 1. The pipeline, step by step: what each step does, how, and what it hands to the next.

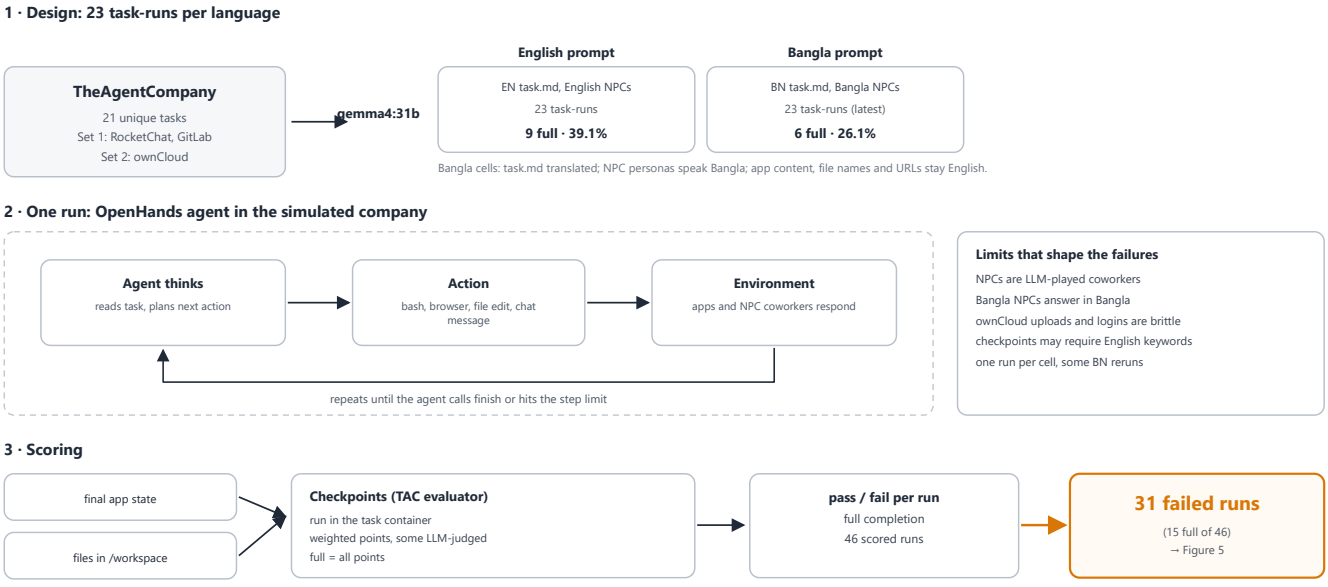


Figure 2. How the study works: the two cells, one run, and the scoring.

R.2 Success rate by run and task set

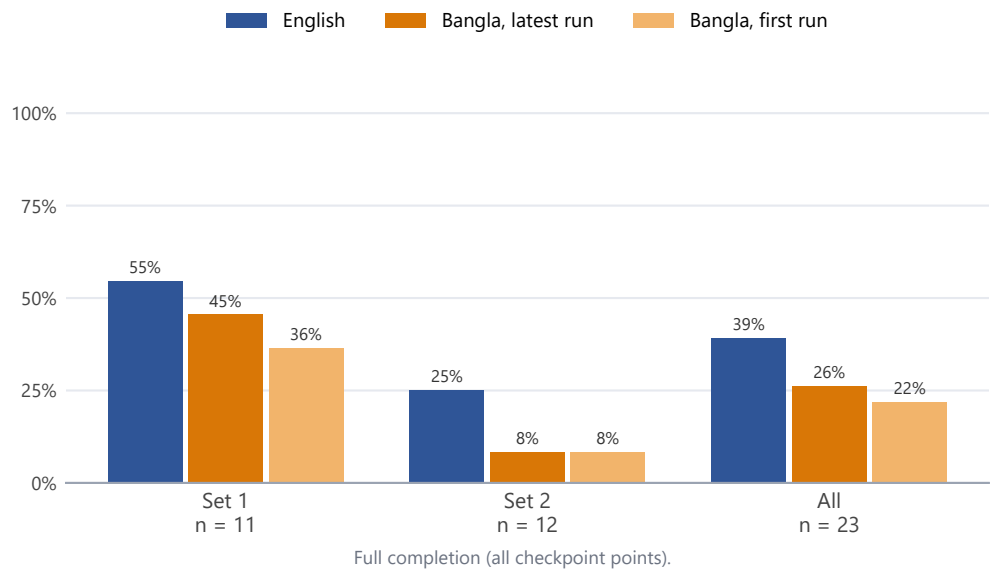


Figure 3. Full-completion rate of English and both Bangla runs, by task set.

- English completes 9 of 23 task-runs, Bangla 6 (latest) or 5 (first run).
- Mean partial score: English 0.515, Bangla 0.433 (latest) / 0.423 (first); Wilcoxon p = 0.141 (latest), 0.046 (first).

R.3 English against Bangla, task by task

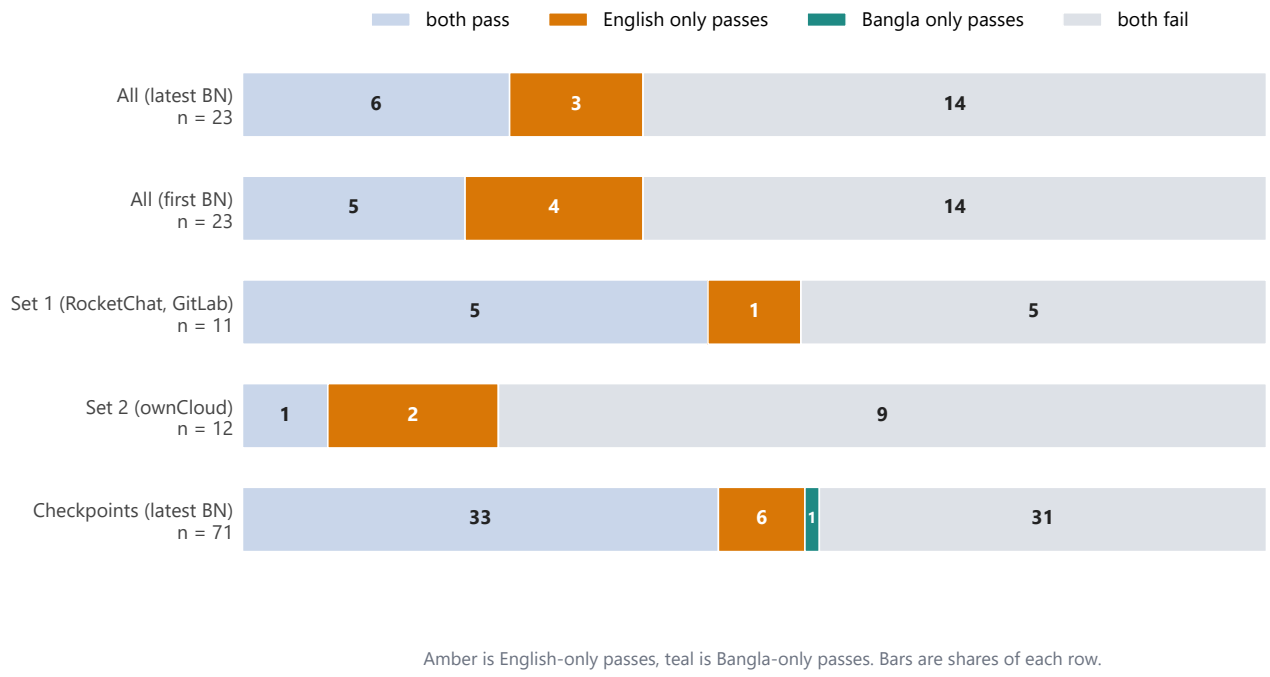


Figure 4. Paired outcome of English and Bangla on the same tasks, and on every checkpoint.

- No task completes only in Bangla: every flip goes English → Bangla (3 latest, 4 first run).
- At checkpoint level 6 checkpoints pass only in English and 1 only in Bangla (McNemar p = 0.125; descriptive, checkpoints are clustered by task).

R.4 Partial score per task

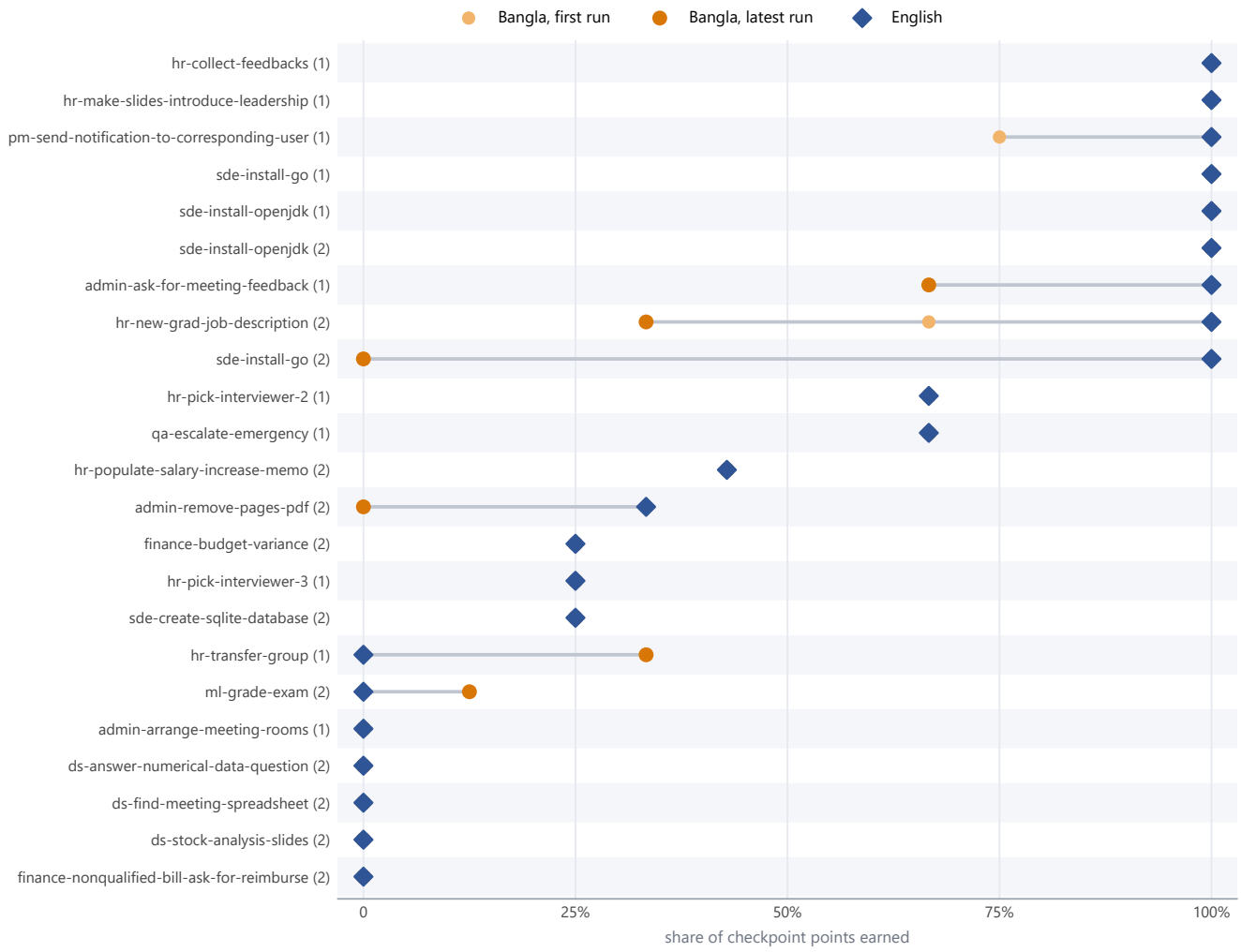


Figure 5. Share of checkpoint points per task: English, latest Bangla run and first Bangla run.

- **17** of 23 task-runs earn exactly the same points in both languages; most failures are shared.
- Where the first and latest Bangla runs differ (light and dark amber apart), the rerun alone moves the score as much as the language does.

R.5 Run-to-run variation

Task	Set	Bangla first run	Bangla rerun	Full completion changed	Score changed
admin-arrange-meeting-rooms	set1	0/2	0/2	no	no
admin-ask-for-meeting-feedback	set1	4/6	4/6	no	no
admin-remove-pages-pdf	set2	0/3	0/3	no	no
ds-answer-numerical-data-question	set2	0/6	0/6	no	no
finance-nonqualified-bill-ask-for-reimburse	set2	0/2	0/2	no	no
hr-collect-feedbacks	set1	5/5	5/5	no	no
hr-make-slides-introduce-leadership	set1	5/5	5/5	no	no
hr-new-grad-job-description	set2	2/3	1/3	no	yes
hr-pick-interviewer-2	set1	4/6	4/6	no	no
hr-pick-interviewer-3	set1	1/4	1/4	no	no
hr-transfer-group	set1	0/3	1/3	no	yes
pm-send-notification-to-corresponding-user	set1	3/4	4/4	**yes**	yes
qa-escalate-emergency	set1	2/3	2/3	no	no

Of 13 Bangla reruns, 1 changed full completion and 3 changed the score. sde-install-go also ran twice in English and in Bangla across the two sets: English passed both times; Bangla passed once and failed once.

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1. Data prepared (pipeline section 1)

Field	Where it comes from
dataset, model, task ID, run ID	TheAgentCompany / gemma4:31b / task name / output folder (`outputs` , `outputs_bn` , `outputs_rc_bn_full` , `outputs_set2*`)
English and Bangla pass/fail	`eval_<task>-image.json` : score = total → full completion; built by `build_tac_runs.py`
Trajectories	`traj_<task>-image.json` ; rendered side by side in `analysis/tac/cases/T0x_*/`
Final state, checkpoints	per-checkpoint points in the eval files; checkpoint definitions in each task's `checkpoints.md`
Bangla task text	`tac-docker/bn/tasks/<task>/task.md`

Alignment: 23 task-runs have both an English and a Bangla result; 2 tasks are English-only and excluded. `old_results/` (runs before the grading and infrastructure fixes) are excluded.

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2. Basic outcome (pipeline section 2)

Setting	n	EN full	BN full	Gap (EN−BN)	MELR	PP	PF	FP	FF
All task-runs, BN latest	23	9 / 23 (39.1%)	6 / 23 (26.1%)	+13.0 pp	33.3%	6	**3**	0	14
All task-runs, BN first	23	9 / 23 (39.1%)	5 / 23 (21.7%)	+17.4 pp	44.4%	5	**4**	0	14
Unique tasks, BN latest	21	7 / 21 (33.3%)	5 / 21 (23.8%)	+9.5 pp	28.6%	5	**2**	0	14
Set 1 (RocketChat, GitLab)	11	6 / 11 (54.5%)	5 / 11 (45.5%)	+9.1 pp	16.7%	5	**1**	0	5
Set 2 (ownCloud)	12	3 / 12 (25.0%)	1 / 12 (8.3%)	+16.7 pp	66.7%	1	**2**	0	9

3. Which cases were analysed (pipeline section 3)

Group	Count	How analysed here
EN-pass / BN-fail (latest)	3	**All 3** read trajectory against trajectory (T01, T02, T03).
Bangla reruns disagree	3	Read (T03, T04, T07): measures run-to-run noise.
EN-fail / BN-fail with different partial score	2	Read (T05, T06).
EN-fail / BN-fail, identical	11	One read as a control (T08); the rest counted only.
EN-pass / BN-pass	6	Counted only.
EN-fail / BN-pass	0	None exist.

4. Trace notes (pipeline section 4)

One note per case with the fields the pipeline asks for (Figure 6 shows the procedure on T01; Figure 7 is the coding scheme). All 8 notes are in Appendix A.

T01 · admin-ask-for-meeting-feedback · C3

- **Task:** admin-ask-for-meeting-feedback
- **Pair class:** EN-pass/BN-fail
- **Checkpoints that differ:** CP5 (2 pts): EN 2/2, both BN runs 0/2. CP1-CP4 identical (all pass) in EN and both BN runs. EN 6/6, BN 4/6 and 4/6.
- **English:** Agent asked Huang Jie for the participant list, then messaged each of the 5 people in English. The NPCs answered in English ('informative and helpful', score 4-5). Agent wrote an xlsx with English feedback, left Chen Xinyi blank. CP5 passed.
- **Bangla:** Agent found the same 5 participants and the same procedure, and wrote the same structure of xlsx (Chen Xinyi blank). The NPCs replied in Bangla (even in the run where the agent's questions were written in English), and the agent stored the Bangla replies verbatim (e.g. 'সহায়ক ছিল'). Scores were all >=4 but the grader's CP5 requires the English substring 'helpful' in each feedback cell, so it failed.
- **First divergence:** NPC replies and therefore the stored Feedback strings are in Bangla; evaluator checks `'helpful' in feedback` (English keyword). Agent behaviour (steps, names, scores, empty row) matches EN.
- **Final failed / missing action:** No agent action was wrong with respect to the Bangla task. Feedback text is not the English word 'helpful' (agent did not translate/normalise feedback to English). The English-only keyword grader is the proximate cause of the 0/2.
- **First failure point:** Unclear · **Type:** Execution/environment failure
- **Confound:** Translation/localization error → C3 (high)
- **Evidence:** Evaluator: all(['helpful' in feedback ...]); BN xlsx Feedback: 'আমার মনে হয় মিটিংটি বেশ তথ্যবহুল এবং সহায়ক ছিল।'

T03 · sde-install-go · C7

- **Task:** sde-install-go
- **Pair class:** run-to-run disagreement
- **Checkpoints that differ:** EN 2/2. BN run A (outputs_bn) 2/2 (identical to EN). BN run B (outputs_set2_bn) 0/2 (CP1 and CP2: `go version` not found on PATH).
- **English:** Agent checked `go version` (not found), wget go1.17.13, `sudo tar -C /usr/local -xzf`, then `export PATH=$PATH:/usr/local/go/bin && go version`, got go1.17.13, finished. Passed.
- **Bangla:** Task is a 3-token translation ('go 1.17 ইনস্টল করুন'). BN run A did exactly the same steps as EN and passed 2/2. BN run B downloaded and untarred without sudo in one command, then verified by calling the absolute path /usr/local/go/bin/go, never exporting PATH, so the grader's `go version` found nothing: 0/2.
- **First divergence:** Step 5-6 of BN run B: verification via absolute path instead of putting go on PATH (EN and BN run A used `export PATH=...`).
- **Final failed / missing action:** Run B: did not make `go` resolvable on PATH (no export / symlink) before finishing.
- **First failure point:** Result verification · **Type:** Verification failure
- **Confound:** Run-to-run noise → C7 (high)
- **Evidence:** Run B: "/usr/local/go/bin/go version" vs run A and EN: "export PATH=$PATH:/usr/local/go/bin && go version"

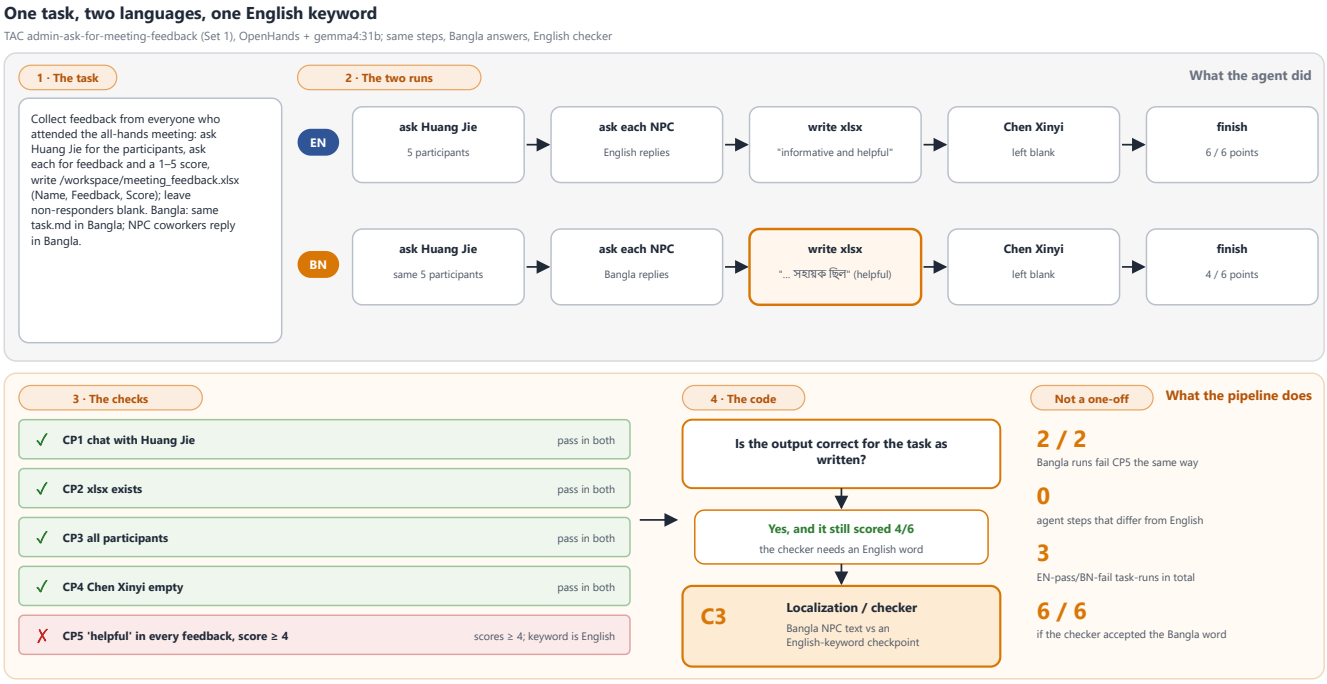


Figure 6. One real task carried through the pipeline: the Bangla answer is right, the checkpoint demands an English keyword.

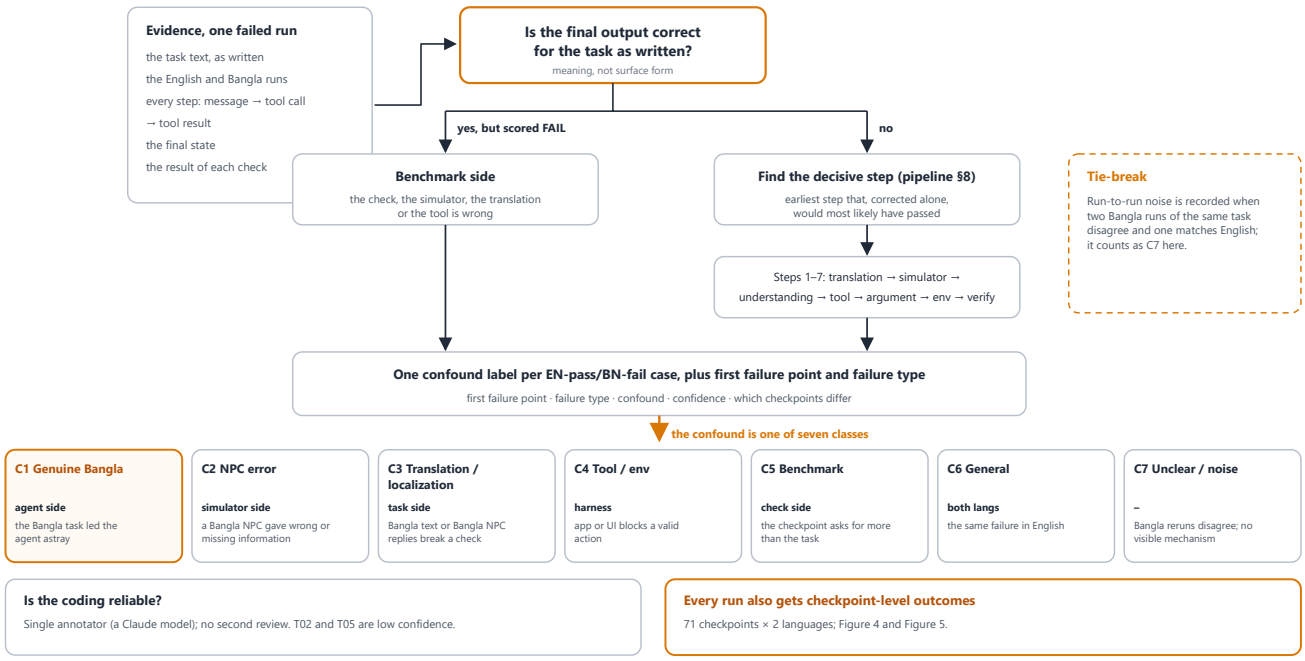


Figure 7. How failures are characterized: one question, one decisive step, one confound class per case.

5–6. First failure point and failure type (pipeline sections 5–6)

Case	Task	Class	First failure point	Failure type	Confound	Confidence
T01	admin-ask-for-meeting-feedback	EN-pass/BN-fail	Unclear	Execution/environment failure	C3	high
T02	hr-new-grad-job-description	EN-pass/BN-fail	Tool-argument construction	Wrong tool argument	C7	low
T03	sde-install-go	run-to-run	Result verification	Verification failure	C7	high
T04	pm-send-notification-to-corresponding-user	run-to-run	Premature termination	Premature abandonment	C7	medium
T05	admin-remove-pages-pdf	EN-fail/BN-fail	Planning	Premature abandonment	C6	low
T06	ml-grade-exam	EN-fail/BN-fail	Result verification	Verification failure	C6	medium
T07	hr-transfer-group	run-to-run	Tool execution	Execution/environment failure	C4	medium
T08	hr-pick-interviewer-2	EN-fail/BN-fail	Planning	Premature abandonment	C6	high

With 8 cases the distributions are not meaningful as shares; the table lists every case. No case starts at instruction understanding or entity grounding: the translated task texts were faithful in all 8 (names, paths and URLs preserved).

7. Confound type (pipeline section 7)

Confound	Cases	Share of 8
C1 Genuine Bangla-related agent failure	0	0%
C2 NPC (user-simulator) error	0	0%
C3 Translation/localization error	1	12%
C4 Tool/environment failure	1	12%
C5 Benchmark/task ambiguity	0	0%
C6 General model weakness present in both languages	3	38%
C7 Unclear (incl. run-to-run noise)	3	38%

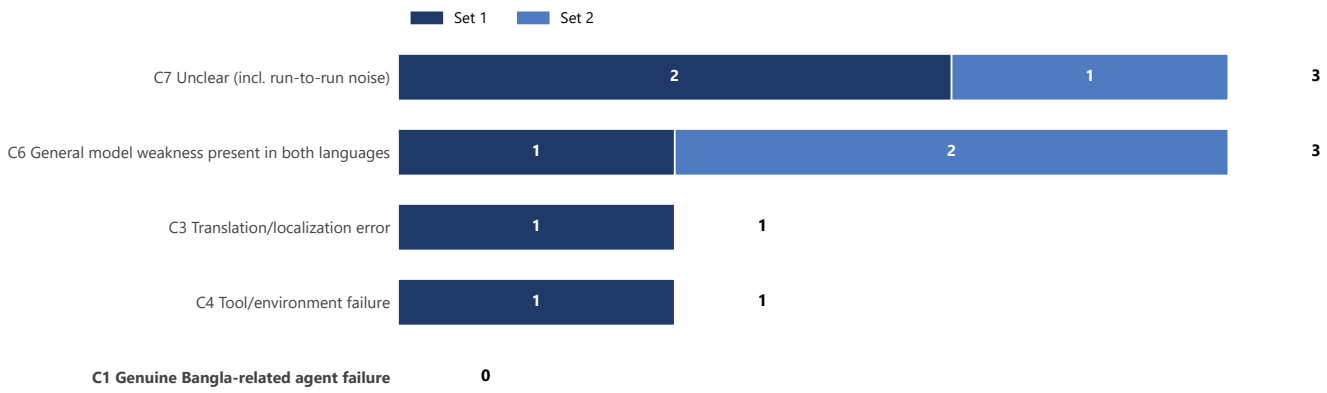


Figure 8. Confound class of the 8 analysed cases, split by task set.

8. Root-cause decision tree (pipeline section 8)

Case	Step 1 translation valid	Step 2 NPC behaved	Step 3 understood	Step 4 right action	Step 5 arguments	Step 6 env executed	Step 7 verified	Exit
T01	ok (faithful)	**NPC replies in Bangla**	ok	ok	ok	ok	ok	Step 0: output right, check rejects it
T02	ok	ok	ok	ok	**unquoted heredoc / wrong folder**	upload wall	–	Step 5 (low confidence, noise)
T03	ok	n/a	ok	ok	ok	ok	**go not on PATH**	Step 7 (noise: run A passed)
T04	ok	ok	ok	ok	ok	ok	**stopped after 2 of 3**	Step 7 (noise: run B passed)
T05	ok	n/a	ok	**no share-link workaround**	–	401 wall	–	Step 4 (EN also failed)
T06	ok	n/a	ok	ok	ok	ok	**missed files, guessed grades**	Step 7 (EN also failed)
T07	ok	n/a	ok	ok	ok	**password prompt, no member list**	–	Step 6
T08	ok	ok	ok	**stopped at first acceptance**	–	–	–	Step 4 (identical in EN)

The pipeline's added Step 0 ("is the output already correct for the task as written?") is the exit for T01: the agent did what the Bangla task asked.

9–10. Evidence and counts (pipeline sections 9–10)

Each note carries the English and Bangla outcome, the differing checkpoints, the decisive step and a quote from the trajectory (Appendix A).

Count	Cases
Genuine Bangla-related agent failures (C1), after filtering	0
NPC (user-simulator)-caused (C2)	0
Translation/localization incl. checker language (C3)	1
Tool/environment-caused (C4)	1
Benchmark/task ambiguity (C5)	0
General weakness present in both (C6)	3
Unclear / run-to-run noise (C7)	3

Dataset-wise and model-wise splits are not applicable (one dataset, one model). Upper bound on genuine Bangla effects: C1 + C7 = 3 cases, all of which are reruns that disagree with each other; lower bound: C1 = 0.

11. Statistical analysis (pipeline section 11)

Paired design. McNemar exact test on discordant task-runs; Wilson intervals for rates; paired bootstrap (4000 resamples, seed 7) for gap and MELR; Wilcoxon signed-rank on partial scores. No Holm correction is needed: a single family, nothing significant.

Setting	EN [95% CI]	BN [95% CI]	Gap [95% CI]	MELR [95% CI]	b / c	McNemar p	Cohen's g
BN latest, 23 task-runs	39.1% [22.2, 59.2]	26.1% [12.5, 46.5]	+13.0 pp [0.0, 26.1]	33.3% [0.0, 66.7]	3 / 0	0.250	0.50
BN first, 23 task-runs	39.1% [22.2, 59.2]	21.7% [9.7, 41.9]	+17.4 pp [4.3, 34.8]	44.4% [11.1, 77.8]	4 / 0	0.125	0.50
BN latest, 21 unique tasks	33.3% [17.2, 54.6]	23.8% [10.6, 45.1]	+9.5 pp [0.0, 23.8]	28.6% [0.0, 66.7]	2 / 0	0.500	0.50

Sensitivity analysis

Scenario	n	EN	BN	Gap	Gap 95% CI (pp)	PF / FP	McNemar p
S0 As scored (BN latest)	23	39.1%	26.1%	+13.0 pp	[0.0, 26.1]	3 / 0	0.250
S1 BN first run instead of latest	23	39.1%	21.7%	+17.4 pp	[4.3, 34.8]	4 / 0	0.125
S2 Unique tasks only	21	33.3%	23.8%	+9.5 pp	[0.0, 23.8]	2 / 0	0.500
S3 Credit the T01 Bangla answers (checker artifact)	23	39.1%	30.4%	+8.7 pp	[0.0, 21.7]	2 / 0	0.500
S4 Drop tasks whose Bangla runs disagree (pm-send, install-go) and T02	19	26.3%	21.1%	+5.3 pp	[0.0, 15.8]	1 / 0	1.000
S5 S3 and S4 together	19	26.3%	26.3%	+0.0 pp	[0.0, 0.0]	0 / 0	1.000

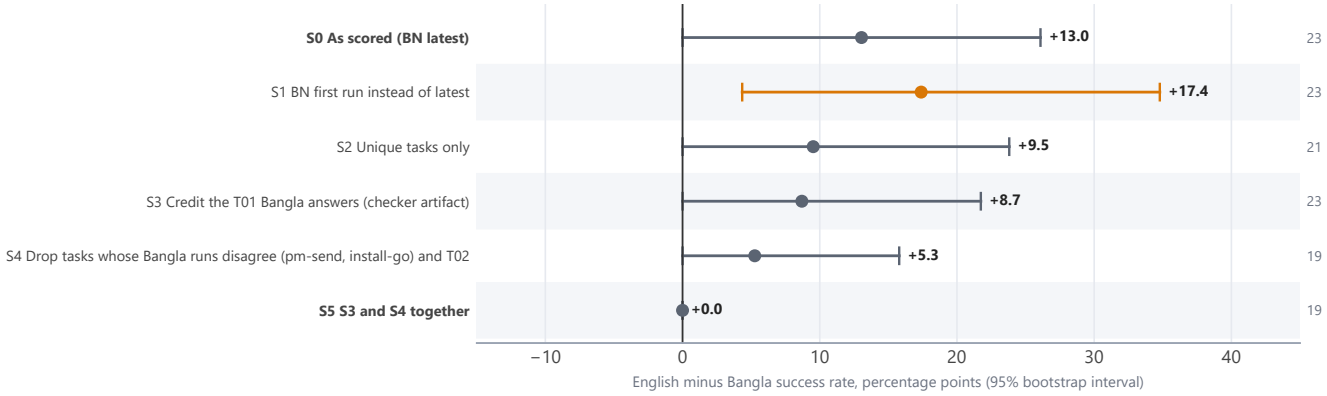


Figure 9. English minus Bangla full-completion gap with 95% interval under each sensitivity scenario. Amber intervals exclude zero.

How to read this. Crediting the one checker artifact and dropping the tasks whose Bangla reruns disagree leaves 0 English-only and 0 Bangla-only task-runs: the gap goes to +0.0 pp. Scenarios S1–S2 show that the choice of rerun and of duplicate tasks moves the estimate by several points on their own.

12. Findings (pipeline section 12)

F1. Direction, not significance. Bangla never completes a task English fails (0 Bangla-only passes) and loses 3–4, but with 23 pairs no test is significant (best p = 0.125). *Boundary:* pilot scale; report as descriptive.

F2. The only stable EN-pass/BN-fail case is a checker artifact. Bangla NPCs answer in Bangla; checkpoint 5 of admin-ask-for-meeting-feedback requires the English word "helpful". The agent's steps match English in both

Bangla runs. *Boundary:* whether Bangla-speaking NPCs are the right design is a benchmark question, not an agent one.

F3. Run-to-run variation is as large as the gap. 1 of 13 Bangla reruns change full completion; sde-install-go passes and fails in Bangla on the same task (the failing run checked go by absolute path and never put it on PATH). *Boundary:* only Bangla was rerun systematically.

F4. Most failures are language-independent. 14 of 23 task-runs fail in both languages and 17 earn identical points; the common causes are ownCloud authentication walls, stopping at the first acceptable answer (hr-pick-interviewer-2) and unverified claims of success (hr-transfer-group). *Boundary:* only 4 both-fail tasks were read.

Not claimed. No case shows the Bangla task text misleading the agent: translation fidelity was checked in all 8 cases.

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13. Paper outputs (pipeline section 13)

#	Output	Status	Where
1	Overall English vs Bangla success table	Done	Section R, Section 2
2	MELR by dataset and model	Done (one dataset, one model, by set)	Section 2
3	Paired outcome table	Done	Section 2, Figure 4
4	Failure-type distribution for PF	Listed per case (n too small for shares)	Sections 5–6
5	First-failure-point distribution	Listed per case	Sections 5–6
6	Confound filtering table	Done	Sections 7, 10
7	Before/after confound-filtered results	Done	Section 11
8	Two or three short case studies	Done	Section 15
9	Appendix with annotation rules and long trajectories	Notes done; rendered trajectories in `analysis/tac/cases/`	Appendix A

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14. Quality-control checklist (pipeline section 14)

Check	Status	Note
Every task ID aligned between English and Bangla	Pass	23 task-runs paired; 2 English-only tasks excluded.
Every metric has a denominator	Pass	Stated in each table.
EN-pass/BN-fail separated from both-fail	Pass	Sections 2–3.
Translation errors not counted as genuine	Pass	C3 is its own class (T01).
User-simulator (NPC) errors counted separately	Pass	C2: 0 cases.
Tool/environment failures counted separately	Pass	C4: T07.
A second reviewer checks a subset	**Not done**	Single annotator; T02 and T05 low confidence.
Disagreements adjudicated	Not done	Depends on the previous item.
Case studies match the quantitative patterns	Pass	T01 (the only stable PF) and T03 (noise).
Strong claims only after confound filtering	Pass	F1–F4 are descriptive and state boundaries.

Other limits: one model; reason for the `*_full` reruns is inferred from folder names; some rendered trajectories are truncated, which limited verification of T02.

15. Case studies

T01 admin-ask-for-meeting-feedback: A correct Bangla answer the checker cannot read. *English:* Agent asked Huang Jie for the participant list, then messaged each of the 5 people in English. The NPCs answered in English ('informative and helpful', score 4-5). Agent wrote an xlsx with English feedback, left Chen Xinyi blank. CP5 passed. *Bangla:* Agent found the same 5 participants and the same procedure, and wrote the same structure of xlsx (Chen Xinyi blank). The NPCs replied in Bangla (even in the run where the agent's questions were written in English), and the agent stored the Bangla replies verbatim (e.g. 'সহায়ক ছিল'). Scores were all >=4 but the grader's CP5 requires the English substring 'helpful' in each feedback cell, so it failed.

Root cause: NPC replies and therefore the stored Feedback strings are in Bangla; evaluator checks `'helpful' in feedback` (English keyword). Agent behaviour (steps, names, scores, empty row) matches EN. *Label:* C3, Translation/localization error, first failure point "Unclear".

T03 sde-install-go: The same task, the same language, two outcomes. *English:* Agent checked `go version` (not found), wget go1.17.13, `sudo tar -C /usr/local -xzf`, then `export PATH=$PATH:/usr/local/go/bin && go version`, got go1.17.13, finished. Passed. *Bangla:* Task is a 3-token translation ('go 1.17 ইনস্টল করুন'). BN run A did exactly the same steps as EN and passed 2/2. BN run B downloaded and untarred without sudo in one command, then verified by calling the absolute path /usr/local/go/bin/go, never exporting PATH, so the grader's `go version` found nothing: 0/2.

Root cause: Step 5-6 of BN run B: verification via absolute path instead of putting go on PATH (EN and BN run A used `export PATH=...`). *Label:* C7, Run-to-run noise, first failure point "Result verification".

T08 hr-pick-interviewer-2: A trap both languages fall into. *English:* Chen Xinyi listed Mark Johnson, Jessica Lee, Emily Zhou and David Wong. Agent messaged Mark, Jessica, Emily in order, Emily agreed, wrote 'Emily Zhou' to interviewer.txt and finished without contacting David Wong (the intentional non-responsive trap). CP2 failed. *Bangla:* Same sequence with Bangla messages: Chen Xinyi's Bangla reply listed the same four names, agent contacted Mark, Jessica, Emily (Mark and Jessica declined, Emily agreed), wrote Emily Zhou, finished without contacting David Wong. CP2 failed identically.

Root cause: None between EN and BN. Both stop after the first acceptance although the task checkpoint expects chatting with all listed interviewers. *Label:* C6, General model weakness (not Bangla-specific), first failure point "Planning".

Appendix A. Trace notes for the 8 analysed cases

T01 · admin-ask-for-meeting-feedback · C3

- **Task:** admin-ask-for-meeting-feedback
- **Pair class:** EN-pass/BN-fail
- **Checkpoints that differ:** CP5 (2 pts): EN 2/2, both BN runs 0/2. CP1-CP4 identical (all pass) in EN and both BN runs. EN 6/6, BN 4/6 and 4/6.
- **English:** Agent asked Huang Jie for the participant list, then messaged each of the 5 people in English. The NPCs answered in English ('informative and helpful', score 4-5). Agent wrote an xlsx with English feedback, left Chen Xinyi blank. CP5 passed.
- **Bangla:** Agent found the same 5 participants and the same procedure, and wrote the same structure of xlsx (Chen Xinyi blank). The NPCs replied in Bangla (even in the run where the agent's questions were written in English), and the agent stored the Bangla replies verbatim (e.g. 'সহায়ক ছিল'). Scores were all >=4 but the grader's CP5 requires the English substring 'helpful' in each feedback cell, so it failed.
- **First divergence:** NPC replies and therefore the stored Feedback strings are in Bangla; evaluator checks `'helpful' in feedback` (English keyword). Agent behaviour (steps, names, scores, empty row) matches EN.
- **Final failed / missing action:** No agent action was wrong with respect to the Bangla task. Feedback text is not the English word 'helpful' (agent did not translate/normalise feedback to English). The English-only keyword grader is the proximate cause of the 0/2.
- **First failure point:** Unclear · **Type:** Execution/environment failure
- **Confound:** Translation/localization error → C3 (high)
- **Evidence:** Evaluator: all(['helpful' in feedback ...]); BN xlsx Feedback: 'আমার মনে হয় মিটিংটি বেশ তথ্যবহুল এবং সহায়ক ছিল।'

T02 · hr-new-grad-job-description · C7

- **Task:** hr-new-grad-job-description
- **Pair class:** EN-pass/BN-fail
- **Checkpoints that differ:** EN 3/3. BN run A (outputs_set2_bn): CP1,CP2 pass, CP3 (content) fail = 2/3. BN run B (outputs_set2_bn_full): CP1 pass, CP2 (file on OwnCloud via link) and CP3 fail = 1/3.
- **English:** Agent merged template.md and requirement.txt with a quoted heredoc (<<'EOF'), uploaded to OwnCloud through the browser after many 'no permission to upload here' attempts, created a public share link, wrote link.txt. All 3 checkpoints passed.
- **Bangla:** The Bangla task text is a faithful translation (names, paths, URL preserved). Run A wrote job_description.md with an UNQUOTED heredoc (<<EOF) so shell expansion could corrupt numerical content (salary; not verifiable because trace truncates the text), uploaded and linked the file, but the LLM content judge (which stresses numerical details) failed it. Run B created the file with the edit tool, hit the same OwnCloud 'no permission to upload' wall as EN, fell back to curl/WebDAV uploads, finally uploaded to the root folder instead of Documents and linked that; CP2 and CP3 failed.
- **First divergence:** Run A: step 4 file creation with unquoted heredoc (EN used quoted 'EOF'). Run B: step ~10-20 OwnCloud upload workaround (WebDAV curl; file ended up in root, not documents).
- **Final failed / missing action:** Run A: likely corrupted/incorrect salary content in job_description.md (cause inferred, not directly visible). Run B: file not placed under documents / link not downloadable by grader.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** Run-to-run noise → C7 (low)
- **Evidence:** Run A: "cat <<EOF > /workspace/job_description.md" (EN: "cat << 'EOF' > ..."); run B finish: 'শেষ পর্যন্ত রুট ডিরেক্টরিতে সফলভাবে আপলোড করেছি'

T03 · sde-install-go · C7

- **Task:** sde-install-go
- **Pair class:** run-to-run disagreement
- **Checkpoints that differ:** EN 2/2. BN run A (outputs_bn) 2/2 (identical to EN). BN run B (outputs_set2_bn) 0/2 (CP1 and CP2: `go version` not found on PATH).
- **English:** Agent checked `go version` (not found), wget go1.17.13, `sudo tar -C /usr/local -xzf`, then `export PATH=$PATH:/usr/local/go/bin && go version`, got go1.17.13, finished. Passed.
- **Bangla:** Task is a 3-token translation ('go 1.17 ইনস্টল করুন'). BN run A did exactly the same steps as EN and passed 2/2. BN run B downloaded and untarred without sudo in one command, then verified by calling the absolute path /usr/local/go/bin/go, never exporting PATH, so the grader's `go version` found nothing: 0/2.
- **First divergence:** Step 5-6 of BN run B: verification via absolute path instead of putting go on PATH (EN and BN run A used `export PATH=...`).
- **Final failed / missing action:** Run B: did not make `go` resolvable on PATH (no export / symlink) before finishing.
- **First failure point:** Result verification · **Type:** Verification failure
- **Confound:** Run-to-run noise → C7 (high)
- **Evidence:** Run B: "/usr/local/go/bin/go version" vs run A and EN: "export PATH=$PATH:/usr/local/go/bin && go version"

T04 · pm-send-notification-to-corresponding-user · C7

- **Task:** pm-send-notification-to-corresponding-user
- **Pair class:** run-to-run disagreement
- **Checkpoints that differ:** EN 4/4. BN run A (outputs_bn) 3/4: CP4 (Mark Johnson notified) fail. BN run B (outputs_rc_bn_full) 4/4.
- **English:** Jessica Lee (NPC) asked in English to notify David Wong, Huang Jie and Mark Johnson. Agent messaged David, Huang, then noticed 'wait, I missed Mark Johnson' and messaged him. 4/4.
- **Bangla:** Jessica replied in Bangla with the three names in Latin script. Run A (3/4): agent translated her reply correctly, informed David Wong and Huang Jie, then called finish without messaging Mark Johnson. Run B (4/4): agent messaged all three (after several wrong-bid retries) and passed.
- **First divergence:** Run A step 17: finish called after 2 of 3 notifications. EN made the same kind of slip but self-corrected. Names were understood correctly in both BN runs.
- **Final failed / missing action:** Run A: no message to Mark Johnson.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** Run-to-run noise → C7 (medium)
- **Evidence:** Run A finish: 'I have informed David Wong and Huang Jie ... as requested by Jessica Lee.' (no Mark Johnson); EN: 'wait, I missed Mark Johnson'

T05 · admin-remove-pages-pdf · C6

- **Task:** admin-remove-pages-pdf
- **Pair class:** EN-fail/BN-fail with differing partial score
- **Checkpoints that differ:** EN 1/3: CP1 (file exists) pass, CP2 (17 pages) and CP3 fail. BN run A and B both 0/3 (no output file).
- **English:** wget of the WebDAV PDF returned 401. After many retries EN created a public share link in OwnCloud and downloaded via <share>/download, then cut the PDF at the page where 'REFERENCES' appears (12 pages, expected 17). Output existed but was wrong (CP1 only).
- **Bangla:** Translation is faithful (References kept, path /workspace/openhands_short.pdf, owncloud URL preserved). In both BN runs wget on the WebDAV and /index.php/f/247 URLs returned 401; the agents never tried the public-share workaround that EN found, declared they could not download without credentials, asked the user for a password, and finished with task_completed=false and no output file.
- **First divergence:** After the first 401 from wget (BN step ~12): EN pivoted to creating a public share link; both BN runs kept retrying authenticated URLs and gave up.
- **Final failed / missing action:** No PDF download, no /workspace/openhands_short.pdf. (EN also failed on content: 12 pages instead of 17.)
- **First failure point:** Planning · **Type:** Premature abandonment
- **Confound:** General model weakness (not Bangla-specific) → C6 (low)
- **Evidence:** BN finish: 'Since I cannot download the file due to authentication requirements ... I am unable to process the PDF' (task_completed=false)

T06 · ml-grade-exam · C6

- **Task:** ml-grade-exam
- **Pair class:** EN-fail/BN-fail with differing partial score
- **Checkpoints that differ:** EN 0/8: CP1 (all 6 attempts downloaded) 0, CP2 (6 gradings) 0/6, CP3 (correct submit) 0. BN 1/8: CP1 0, CP2 1/6 (Attempt1 scored 1, which matches the key), CP3 0. Only one BN run available.
- **English:** Agent explored OwnCloud folders, found 5 of the 6 attempt PDFs (missed Attempt6), downloaded them with credentials, assigned scores 0/10/8/7/5 (none match the key) and named Attempt2_10-submit.
- **Bangla:** Faithful translation. Agent found only 3 of 6 attempts (1, 2, 5), initially downloaded empty/HTML stubs (412 B) after 401s, then recovered with credentials from /npc/npc_credential.json, scored 1/13/12 (Attempt1_1 exactly correct, others wrong) and chose Attempt2_13-submit.
- **First divergence:** Search phase: neither run enumerated all six locations; BN stopped earlier (3 files vs 5). Grading phase: both graded by guessing, EN 0/5 correct, BN 1/3 correct.
- **Final failed / missing action:** Missed Attempts 3, 4, 6 (BN) / Attempt 6 (EN); wrong scores; wrong submit file in both (correct is Attempt2_15).
- **First failure point:** Result verification · **Type:** Verification failure
- **Confound:** General model weakness (not Bangla-specific) → C6 (medium)
- **Evidence:** BN final files: Attempt1_1.pdf, Attempt2_13-submit.pdf, Attempt5_12.pdf (expected Attempt2_15-submit, 6 files)

T07 · hr-transfer-group · C4

- **Task:** hr-transfer-group
- **Pair class:** run-to-run disagreement
- **Checkpoints that differ:** EN 0/3 (all fail). BN run A (outputs_bn) 0/3. BN run B (outputs_rc_bn_full) 1/3: CP2 (Li Ming added to #project-ai) pass; CP1 (removed from #project-graphdb) and CP3 (bio) fail.
- **English:** Agent used Rocket.Chat admin panel (Rooms/Users) and then finished claiming Li Ming was moved and bio updated; the grader shows none of the three changes took effect (0/3), i.e. unverified false success.
- **Bangla:** Translation is faithful (channel names, user name preserved). Run A: bio edit in admin Users page was blocked by a password prompt, Rooms admin view had no member list, agent finished with task_completed=false (0/3). Run B: used the channel Members panel, saw 'No members found' in #project-graphdb, added Li Ming to #project-ai (1/3), bio save again hit the password prompt, finished claiming success.
- **First divergence:** Run B chose the channel Members UI and succeeded at the add; run A and EN took the admin-Rooms route which did not change channel membership. Both BN runs hit the admin password prompt on bio save.
- **Final failed / missing action:** Bio not saved (password modal), Li Ming not removed from #project-graphdb (member list empty/unavailable in UI) in every run; EN additionally did not add him to #project-ai.
- **First failure point:** Tool execution · **Type:** Execution/environment failure
- **Confound:** Tool/environment failure → C4 (medium)
- **Evidence:** BN: 'when attempting to save, the system requested a password for security, which I do not have.' BN run B score 1/3 vs run A 0/3, EN 0/3

T08 · hr-pick-interviewer-2 · C6

- **Task:** hr-pick-interviewer-2
- **Pair class:** EN-fail/BN-fail with identical partial score (4/6 both; no differing score)
- **Checkpoints that differ:** None. EN and BN both 4/6: CP1 2/2, CP2 0/2, CP3 2/2.
- **English:** Chen Xinyi listed Mark Johnson, Jessica Lee, Emily Zhou and David Wong. Agent messaged Mark, Jessica, Emily in order, Emily agreed, wrote 'Emily Zhou' to interviewer.txt and finished without contacting David Wong (the intentional non-responsive trap). CP2 failed.
- **Bangla:** Same sequence with Bangla messages: Chen Xinyi's Bangla reply listed the same four names, agent contacted Mark, Jessica, Emily (Mark and Jessica declined, Emily agreed), wrote Emily Zhou, finished without contacting David Wong. CP2 failed identically.
- **First divergence:** None between EN and BN. Both stop after the first acceptance although the task checkpoint expects chatting with all listed interviewers.
- **Final failed / missing action:** No message to David Wong in either condition.
- **First failure point:** Planning · **Type:** Premature abandonment
- **Confound:** General model weakness (not Bangla-specific) → C6 (high)
- **Evidence:** BN finish lists Mark, Jessica, Emily only; EN finish: 'Emily Zhou agreed ... I have written her name' (no David Wong)

Appendix B. Every paired task-run

Task	Set	EN points	BN latest	BN first	Outcome (latest)	BN latest run folder
admin-arrange-meeting-rooms	set1	0/2	0/2	0/2	FF	outputs_rc_bn_full
admin-remove-pages-pdf	set2	1/3	0/3	0/3	FF	outputs_set2_bn_full
ds-answer-numerical-data-question	set2	0/6	0/6	0/6	FF	outputs_set2_bn_full
ds-find-meeting-spreadsheet	set2	0/2	0/2	0/2	FF	outputs_set2_bn
ds-stock-analysis-slides	set2	0/8	0/8	0/8	FF	outputs_set2_bn
finance-budget-variance	set2	1/4	1/4	1/4	FF	outputs_set2_bn
finance-nonqualified-bill-ask-for-reimburse	set2	0/2	0/2	0/2	FF	outputs_set2_bn_full
hr-pick-interviewer-2	set1	4/6	4/6	4/6	FF	outputs_rc_bn_full
hr-pick-interviewer-3	set1	1/4	1/4	1/4	FF	outputs_rc_bn_full
hr-populate-salary-increase-memo	set2	3/7	3/7	3/7	FF	outputs_set2_bn
hr-transfer-group	set1	0/3	1/3	0/3	FF	outputs_rc_bn_full
ml-grade-exam	set2	0/8	1/8	1/8	FF	outputs_set2_bn
qa-escalate-emergency	set1	2/3	2/3	2/3	FF	outputs_rc_bn_full
sde-create-sqlite-database	set2	2/8	2/8	2/8	FF	outputs_set2_bn
admin-ask-for-meeting-feedback	set1	6/6	4/6	4/6	PF	outputs_rc_bn_full
hr-new-grad-job-description	set2	3/3	1/3	2/3	PF	outputs_set2_bn_full
sde-install-go	set2	2/2	0/2	0/2	PF	outputs_set2_bn
hr-collect-feedbacks	set1	5/5	5/5	5/5	PP	outputs_rc_bn_full
hr-make-slides-introduce-leadership	set1	5/5	5/5	5/5	PP	outputs_rc_bn_full
pm-send-notification-to-corresponding-user	set1	4/4	4/4	3/4	PP	outputs_rc_bn_full
sde-install-go	set1	2/2	2/2	2/2	PP	outputs_bn
sde-install-openjdk	set1	2/2	2/2	2/2	PP	outputs_bn
sde-install-openjdk	set2	2/2	2/2	2/2	PP	outputs_set2_bn